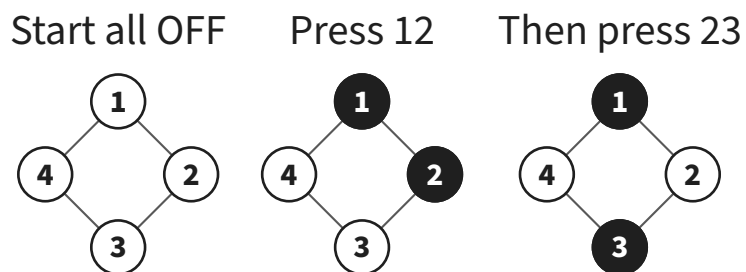
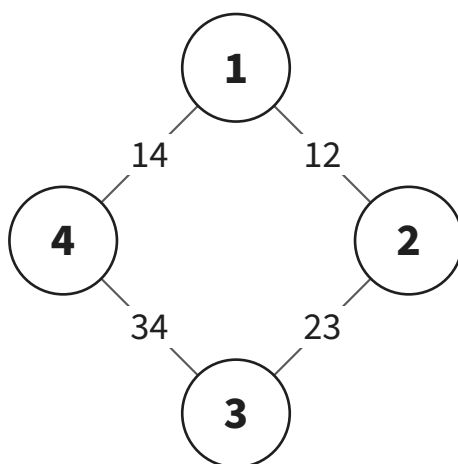


Problem 1: Put a two-sided counter on each big lamp, white side up: all OFF. Point to a line. Flip the counters at *both* ends. Dark means ON.

The line joining lamps 1 and 2 is called **12**. Replay these two moves on your big board.



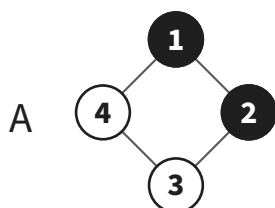
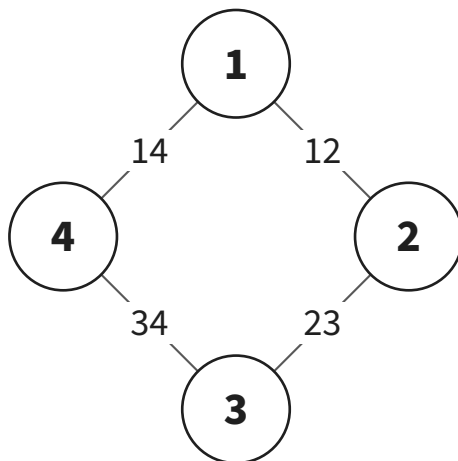
Write the moves in order: **12, 23**.



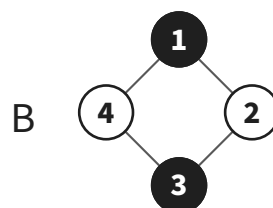
Make the dark lamps white again by flipping pairs. Write your moves so a partner can replay them.

Moves: _____

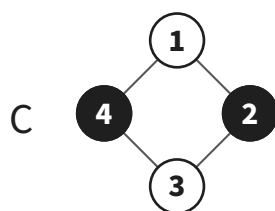
Problem 2: Make each target picture on the big board. Reset all counters to white before each target. Flip both ends of one line at a time. Write your moves in order.



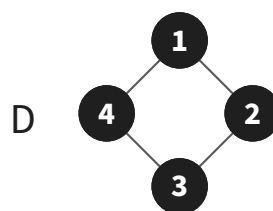
Moves: _____



Moves: _____



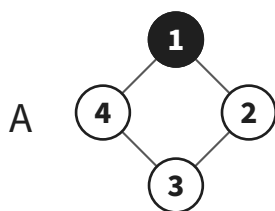
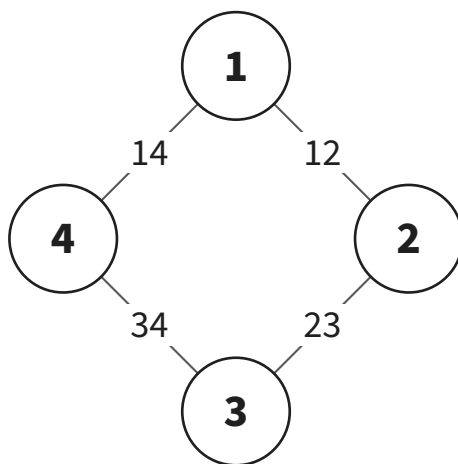
Moves: _____



Moves: _____

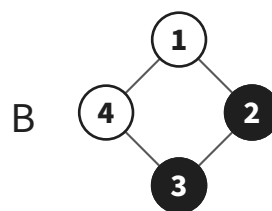
Give a partner one of your move lists. Reset all OFF. Have your partner replay it and point to its target.

Problem 3: Try each target picture. Start all OFF each time. Flip both ends of one line at a time. Write a move list that works, or check “not yet.”



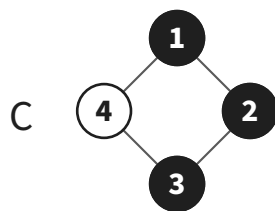
Moves: _____

☐ Not yet



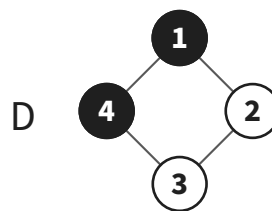
Moves: _____

☐ Not yet



Moves: _____

☐ Not yet

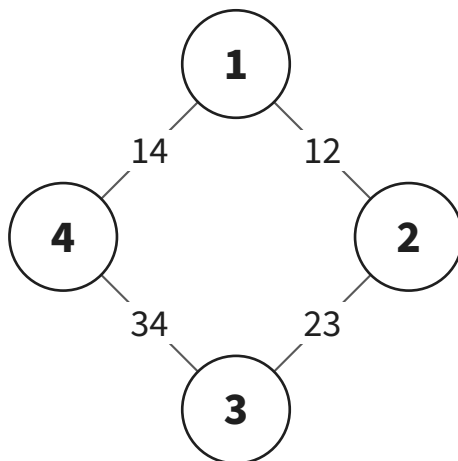


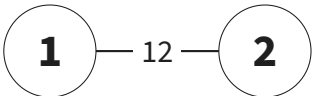
Moves: _____

☐ Not yet

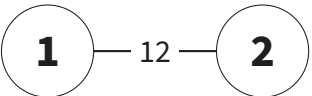
For a “not yet” picture, try a different move list on the board. Keep any list that works.

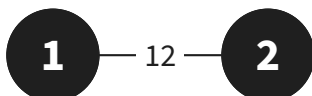
Problem 4: For each row, turn counters over by hand to set up its starting picture on the big board. Leave lamps 3 and 4 OFF. Then press 12 once. Darken the ON lamps in the answer picture. Count ON lamps before and after.



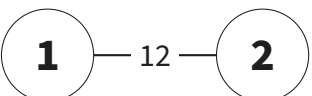
A 
ON before: _____

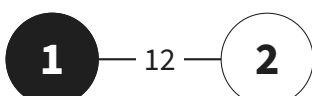
press 12 →


ON after: _____

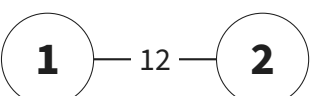
B 
ON before: _____

press 12 →

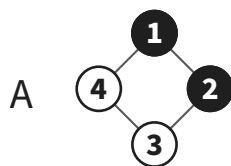

ON after: _____

C 
ON before: _____

press 12 →

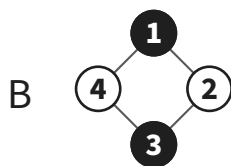

ON after: _____

Problem 5: These are the targets you tried. For each picture, circle “made” or “not yet.” Count its ON lamps. Use the big board to try any picture again; reset all OFF each time.



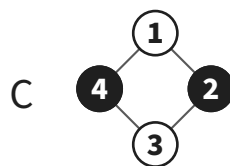
ON: _____

made / not yet



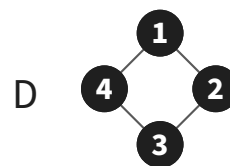
ON: _____

made / not yet



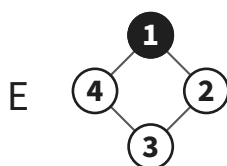
ON: _____

made / not yet



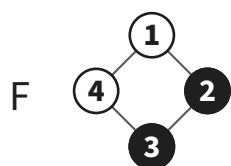
ON: _____

made / not yet



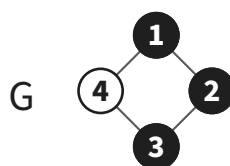
ON: _____

made / not yet



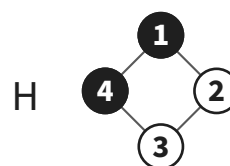
ON: _____

made / not yet



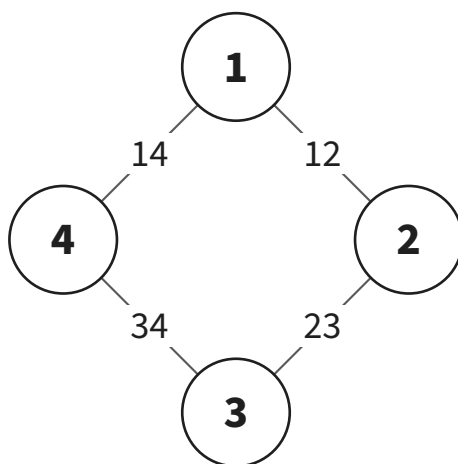
ON: _____

made / not yet



ON: _____

made / not yet

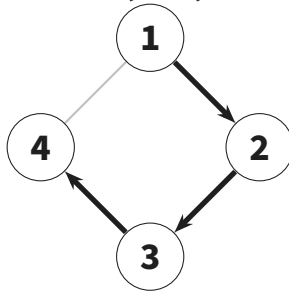


Compare the ON counts in the pictures you made.

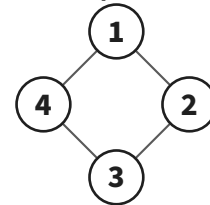
Point to what is alike. Then compare the “not yet” pictures.

Problem 6: Follow each arrow path on the big board: flip both ends of each line, in order. Start all OFF for each path. Darken the ON lamps in its final picture.

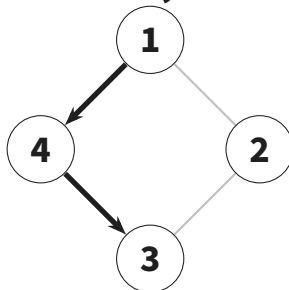
A: 12, 23, 34



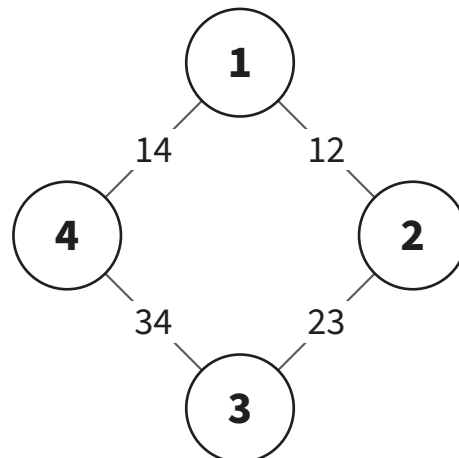
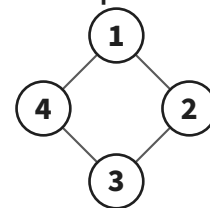
Final picture:



B: 14, 43

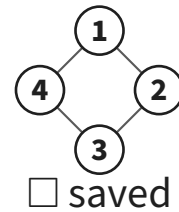
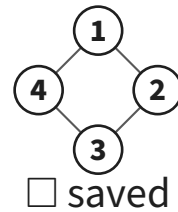
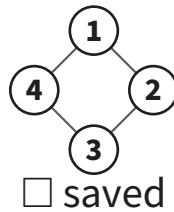
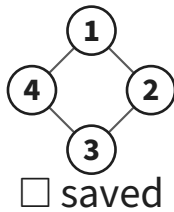
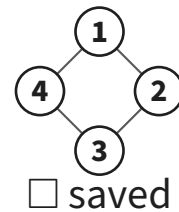
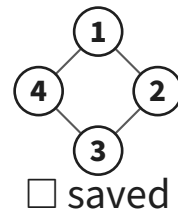
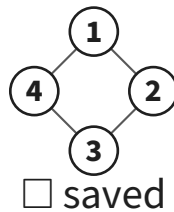
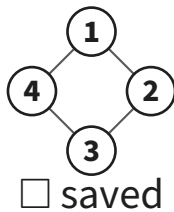
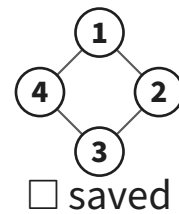
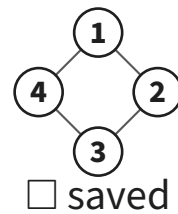
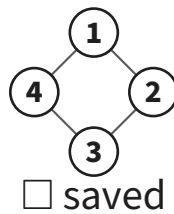
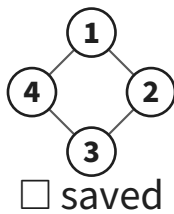
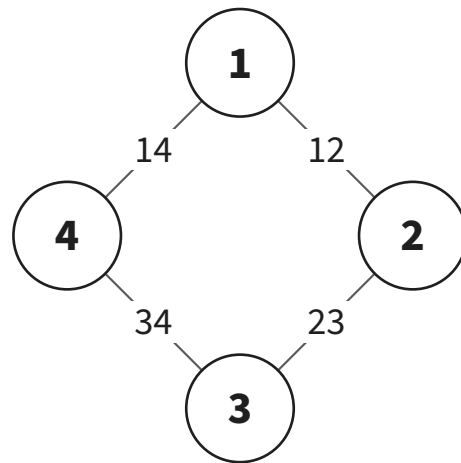


Final picture:



Replay one path slowly. Point to each lamp that gets flipped twice. Point to each lamp flipped just once.

Problem 7: Start all OFF on the big board. Try some moves. Copy the result onto a little ring: darken its ON lamps. Reset and make different pictures. Include all OFF. Leave extra rings empty.



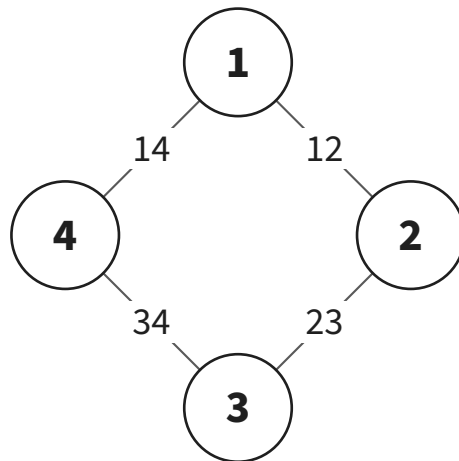
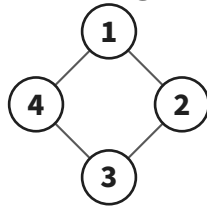
Check “saved” under each picture you record, including all OFF. This keeps unused rings separate.

Problem 8: Look at the pictures you saved in Problem 7. Count the ON lamps in each. Write that count beside its little ring. Point to the pictures with the same count. Count how many different pictures you have.

I saved _____ different pictures.

Choose a picture whose moves you have not written down. Copy it here. Start all OFF on the big board and find a move list that makes it.

My target:



My moves: _____

Reset all OFF and ask a partner to replay your list. Compare the result with your target.